

PhD Course Macroeconomics I: Sustainability

Problem 2: The Optimal Carbon Tax: Derivation and Sensitivity Analysis

Jonathan Thiel, Zsigmond Szajbely, Anushka Mukherjee

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Part 1 – Analytical Derivation

Setup of the model (Section 25.4.2)

Preferences of the representative consumer in the oil-consuming region are

$$\sum_{t=0}^{\infty} \beta^t \log C_t. \quad (25.20)$$

Final-good production is Cobb–Douglas with carbon damages entering through TFP:

$$Y_t = \underbrace{\exp(z_t - \gamma S_t)}_{\equiv A_t} L_t^{1-\alpha-\nu} K_t^\alpha E_t^\nu. \quad (25.21)$$

Energy services aggregate three fuels (oil o , coal c , green g) via CES:

$$E_t = (\lambda_o e_{o,t}^\rho + \lambda_c e_{c,t}^\rho + \lambda_g e_{g,t}^\rho)^{1/\rho}, \quad \lambda_o + \lambda_c + \lambda_g = 1. \quad (25.22)$$

Total carbon emissions in period t are $M_t = e_{o,t} + e_{c,t}$ (eq. 25.24), so the atmospheric excess carbon stock follows

$$S_t - S_0 = \sum_{s=0}^t (1 - d_s) M_{t-s}, \quad (25.9)$$

where $1 - d_s$ is the share of a unit emitted s periods ago that is still in the atmosphere.

Conventional oil is in finite supply R_t , extracted at zero cost:

$$R_{t+1} = R_t - e_{o,t}, \quad R_t \geq 0. \quad (25.25)$$

For the social planner who internalises both regions, oil extraction is costless, and coal/green require $p_{\kappa,t}$ units of final good per unit of energy ($\kappa \in \{c, g\}$). The economy-wide resource constraint is therefore

$$C_t + K_{t+1} + p_{c,t} e_{c,t} + p_{g,t} e_{g,t} = Y_t, \quad (25.23')$$

(this is (25.23) without the international payment $p_{o,t}e_{o,t}$, which is internal to the joint planner).

(a) Lagrangian and First-Order Conditions

The planner solves

$$\max_{\{C_t, K_{t+1}, e_{o,t}, e_{c,t}, e_{g,t}, R_{t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \log C_t \quad \text{s.t. (25.9), (25.21), (25.23'), (25.25).}$$

Attach multipliers $\beta^t \mu_t$ to the resource constraint, $\beta^t \eta_t$ to the definition of S_t , and $\beta^t \xi_t$ to the oil-stock law of motion. The (current-value) Lagrangian reads

$$\begin{aligned} \mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big\{ & \log C_t + \mu_t [e^{z_t - \gamma S_t} L_t^{1-\alpha-\nu} K_t^\alpha E_t^\nu - p_{c,t} e_{c,t} - p_{g,t} e_{g,t} - C_t - K_{t+1}] \\ & + \eta_t \left[S_0 + \sum_{s=0}^t (1-d_s)(e_{o,t-s} + e_{c,t-s}) - S_t \right] \\ & + \xi_t [R_t - e_{o,t} - R_{t+1}] \Big\}. \end{aligned}$$

Useful partial derivatives. From (25.21) and (25.22),

$$\frac{\partial Y_t}{\partial K_t} = \alpha \frac{Y_t}{K_t}, \quad \frac{\partial Y_t}{\partial S_t} = -\gamma Y_t, \quad \frac{\partial Y_t}{\partial e_{\kappa,t}} = \nu \frac{Y_t}{E_t^\rho} \lambda_\kappa e_{\kappa,t}^{\rho-1}.$$

FOC w.r.t. C_t

$$\frac{1}{C_t} = \mu_t. \tag{A.1}$$

FOC w.r.t. K_{t+1}

$$-\beta^t \mu_t + \beta^{t+1} \mathbb{E}_t \left[\mu_{t+1} \alpha \frac{Y_{t+1}}{K_{t+1}} \right] = 0,$$

which, using (A.1), yields the Euler equation

$$\frac{1}{C_t} = \beta \mathbb{E}_t \left[\frac{1}{C_{t+1}} \cdot \alpha \frac{Y_{t+1}}{K_{t+1}} \right]. \tag{A.2}$$

FOC w.r.t. $e_{o,t}$ (oil)

Oil enters Y_t (through E_t), emissions M_t , and the oil stock. A unit of $e_{o,t}$ shifts S_{t+j} by $(1-d_j)$ for every $j \geq 0$, so

$$\mu_t \nu \frac{Y_t}{E_t^\rho} \lambda_o e_{o,t}^{\rho-1} + \sum_{j=0}^{\infty} \beta^j \mathbb{E}_t \eta_{t+j} (1-d_j) - \xi_t = 0. \tag{A.3}$$

FOC w.r.t. $e_{c,t}$ (coal)

Coal has a marginal production cost $p_{c,t}$ in final-good units and also emits carbon:

$$\mu_t \nu \frac{Y_t}{E_t^\rho} \lambda_c e_{c,t}^{\rho-1} - \mu_t p_{c,t} + \sum_{j=0}^{\infty} \beta^j \mathbb{E}_t \eta_{t+j} (1 - d_j) = 0. \quad (\text{A.4})$$

FOC w.r.t. $e_{g,t}$ (green)

Green energy costs $p_{g,t}$ but causes *no* emissions:

$$\mu_t \nu \frac{Y_t}{E_t^\rho} \lambda_g e_{g,t}^{\rho-1} = \mu_t p_{g,t} \iff \nu \frac{Y_t}{E_t^\rho} \lambda_g e_{g,t}^{\rho-1} = p_{g,t}. \quad (\text{A.5})$$

FOC w.r.t. S_t

$$\mu_t (-\gamma Y_t) - \eta_t = 0 \implies \eta_t = -\gamma \mu_t Y_t = -\gamma \frac{Y_t}{C_t}. \quad (\text{A.6})$$

$\eta_t < 0$: extra atmospheric carbon is a “bad”.

FOC w.r.t. R_{t+1} (Hotelling rule)

R_{t+1} enters period t with $-\xi_t$ and period $t+1$ with $+\xi_{t+1}$, so

$$-\beta^t \xi_t + \beta^{t+1} \mathbb{E}_t \xi_{t+1} = 0 \implies \xi_t = \beta \mathbb{E}_t \xi_{t+1}. \quad (\text{A.7})$$

The shadow value of in-ground oil grows at rate $1/\beta$.

Difference between oil and coal FOCs: implicit Pigou wedge

Subtracting (A.4) from (A.3), and using $\mu_t = 1/C_t$ and (A.6),

$$\underbrace{\mu_t \nu \frac{Y_t}{E_t^\rho} (\lambda_o e_{o,t}^{\rho-1} - \lambda_c e_{c,t}^{\rho-1})}_{\text{net marg. productivity gap of oil over coal}} + \mu_t p_{c,t} = \xi_t.$$

Equivalently, rearranging (A.4) gives the optimal marginal energy condition with carbon wedge:

$$\underbrace{\nu \frac{Y_t}{E_t^\rho} \lambda_c e_{c,t}^{\rho-1}}_{\text{MPE of coal}} = \underbrace{p_{c,t}}_{\text{prod. cost}} + \underbrace{C_t \sum_{j=0}^{\infty} \beta^j \mathbb{E}_t \frac{\gamma(1-d_j)Y_{t+j}}{C_{t+j}}}_{\text{carbon damage} \equiv \tau_t}. \quad (\text{A.4}')$$

The bracketed sum is precisely the marginal social damage of one extra unit of carbon emitted at t . We isolate it in part (b).

(b) The Pigou Carbon-Tax Formula (eq. 25.32)

Pigou principle. The optimal carbon tax equals the social marginal damage of a unit emission, expressed in current consumption units:

$$\tau_t = \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j \frac{U'(C_{t+j})}{U'(C_t)} \cdot \left(-\frac{\partial F_{t+j}}{\partial S_{t+j}} \right) \cdot \frac{\partial S_{t+j}}{\partial e_{\kappa,t}}. \quad (25.30)$$

We now evaluate each factor under the functional forms of Section 25.4.2.

Step 1: Marginal damage flow. With $Y_t = e^{-\gamma S_t} \tilde{Y}_t$ (where $\tilde{Y}_t$ collects everything other than the damage factor),

$$\frac{1}{Y_t} \frac{\partial Y_t}{\partial S_t} = \frac{1}{e^{-\gamma S_t} \tilde{Y}_t} \cdot (-\gamma) e^{-\gamma S_t} \tilde{Y}_t = -\gamma. \quad (25.31)$$

Hence $-\partial F_{t+j}/\partial S_{t+j} = \gamma Y_{t+j}$: a constant fraction γ of GDP is lost per extra unit of S .

Step 2: Marginal utility ratio. Logarithmic utility implies $U'(C) = 1/C$, so

$$\frac{U'(C_{t+j})}{U'(C_t)} = \frac{C_t}{C_{t+j}}.$$

Step 3: Carbon-cycle response. From (25.9), one unit of $e_{\kappa,t}$ ($\kappa \in \{o, c\}$) raises S_{t+j} by $(1 - d_j)$:

$$\frac{\partial S_{t+j}}{\partial e_{\kappa,t}} = 1 - d_j.$$

Step 4: Assemble. Substituting into (25.30):

$$\tau_t = \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j \frac{C_t}{C_{t+j}} \gamma Y_{t+j} (1 - d_j) = C_t \gamma \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j (1 - d_j) \frac{Y_{t+j}}{C_{t+j}}. \quad (*)$$

Step 5: Constant savings/consumption ratio. The combination of log-utility, Cobb–Douglas production, full depreciation and the within-period lump-sum rebate of carbon-tax revenue implies a *constant* aggregate savings rate

$$s = \frac{\alpha\beta}{1-\nu} \implies C_t = (1-s)(1-\nu)Y_t \equiv \theta Y_t,$$

with $\theta \equiv (1-s)(1-\nu)$ independent of t . Therefore

$$\frac{Y_{t+j}}{C_{t+j}} = \frac{1}{\theta} = \frac{Y_t}{C_t} \quad \text{for all } j.$$

Step 6: Pulling Y_t/C_t out and cancelling C_t . Plugging the constancy of Y/C into (*):

$$\tau_t = C_t \gamma \frac{Y_t}{C_t} \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j (1 - d_j) = Y_t \gamma \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j (1 - d_j).$$

Hence

$$\tau_t = Y_t \cdot \gamma \sum_{j=0}^{\infty} \beta^j (1 - d_j), \quad (25.32)$$

which is exactly the textbook formula.

Interpretation – the three “d”s.

- *Decay d_j* : The longer carbon lingers in the atmosphere (high $1 - d_j$), the larger the integrated damage and the higher τ_t .
- *Damage parameter γ* : Tax scales linearly with the marginal damage flow.
- *Discount factor β* : A higher β weights future losses more heavily, raising τ_t .

Remarkably, the optimal tax-to-GDP ratio depends only on these three primitives: *not* on the energy mix $\{\lambda_o, \lambda_c, \lambda_g\}$, the elasticity of substitution ρ , capital share α , energy share ν , or the stock of oil R_t . This is the celebrated Golosov–Hassler–Krusell–Tsyvinski (2014) closed-form result.

Part 2 — Numerical Replication and Sensitivity Analysis

We implement the dynamic two-region IAM of Golosov et al. (2014), as presented in Chapter 25 of Azzimonti et al. (2025). The economy features a representative household, a final-goods sector with a CES energy aggregator over oil, coal, and green energy, a finite oil endowment governed by the Hotelling rule under log utility, a two-component carbon cycle, and an ECS-based temperature equation. All simulations are carried out in Python; the full code is submitted alongside this document.

Model summary. Production combines capital K_t , labour $L = 1$, and aggregate energy E_t :

$$Y_t = A_t K_t^\alpha E_t^\nu, \quad A_t = \exp(gt - \gamma S_t), \quad (1)$$

where g is the TFP growth rate, γ is the Nordhaus damage parameter, and S_t is atmospheric excess carbon (GtC above pre-industrial). The CES energy aggregator is

$$E_t = [\lambda_o e_{o,t}^\rho + \lambda_c e_{c,t}^\rho + \lambda_g e_{g,t}^\rho]^{1/\rho}. \quad (2)$$

Carbon stock evolves via the Golosov two-component cycle:

$$S_{t+1}^P = S_t^P + \phi_L M_t, \quad S_{t+1}^T = (1 - d_T) S_t^T + (1 - \phi_L) M_t, \quad (3)$$

where $M_t = \kappa_o e_{o,t} + \kappa_c e_{c,t}$ are total fossil fuel emissions, $\phi_L = 0.20$ is the permanent atmospheric fraction, and $d_T = 1 - (1 - 0.0228)^{10}$ is the per-decade transient decay rate. Temperature follows from equilibrium climate sensitivity:

$$T_t = \text{ECS} \cdot \frac{\ln((S_t^P + S_t^T + S_{\text{pre}})/S_{\text{pre}})}{\ln 2}. \quad (4)$$

The optimal carbon tax (Goloso formula, eq. 25.32) is

$$\tau_t^* = \gamma Y_t \left[\frac{\phi_L}{1 - \beta} + \frac{1 - \phi_L}{1 - \beta(1 - d_T)} \right]. \quad (5)$$

Calibration follows Goloso et al. (2014): $\alpha = 0.30$, $\nu = 0.05$, $\beta = 0.985^{10}$, $\rho = 0.5$ (EoS= 2), $\gamma = 5.3 \times 10^{-5}$ GtC⁻¹, ECS = 3 °C, and initial atmospheric carbon $S_{2020} = 893.5$ GtC. The emission scale factor ξ is calibrated to match initial fossil fuel emissions of ≈ 100 GtC per decade. The initial capital stock is set at its balanced growth path value, $K_0 \approx 0.097$.

(a) Replication of Figure 25.6

Figure 1 shows simulated global temperature under three policy regimes over 2020–2200: no tax, the Goloso optimal tax τ^* (eq. 5), and a moderate tax of $\frac{1}{3}\tau^*$. Table 1 summarises the temperature endpoints.

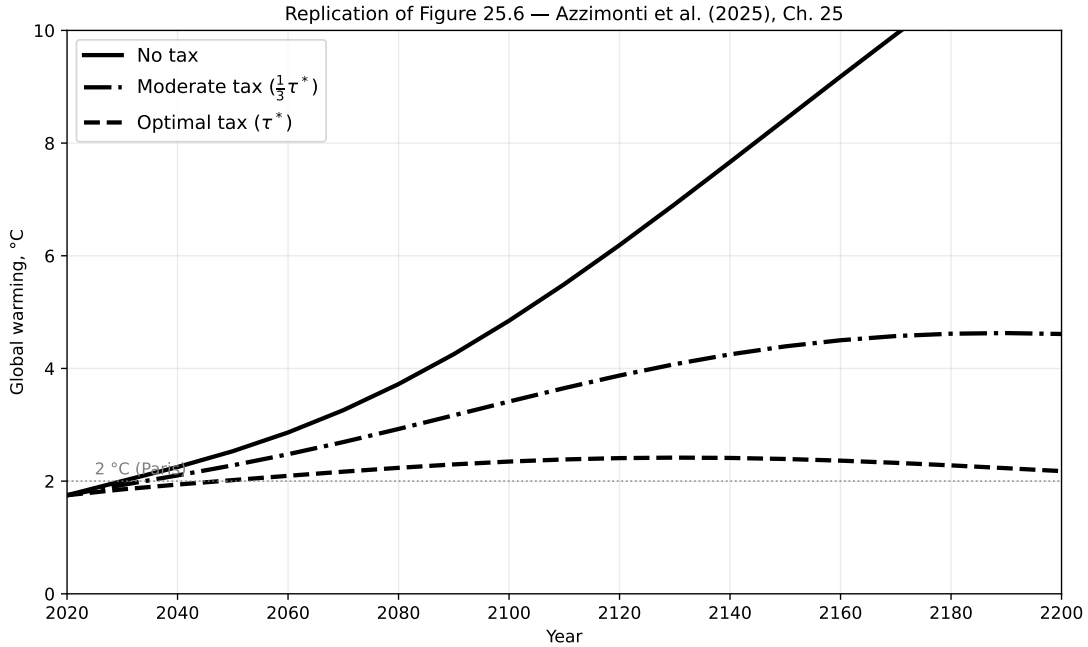


Figure 1: Replication of Figure 25.6 from Azzimonti et al. (2025). Global mean temperature above pre-industrial levels under three carbon tax scenarios, 2020–2200. The dashed horizontal line marks the 2 °C Paris Agreement target.

Policy	T_0 (°C)	T_{2100} (°C)	T_{2200} (°C)
No tax	1.75	4.85	12.04
Moderate ($\frac{1}{3}\tau^*$)	1.75	3.41	4.61
Optimal (τ^*)	1.75	2.35	2.18

Table 1: Temperature at key years under each policy.

Discrepancies. The textbook figure reports approximately 9 °C by 2200 under no policy, whereas our replication yields 12 °C. Two sources account for this. First, our initial temperature (1.75 °C) exceeds the observed 1.1 °C because we apply equilibrium climate sensitivity directly to current CO₂ concentrations; the

true transient response is lower than ECS because ocean heat uptake delays warming. Second, we calibrate the green energy price ($p_g = 2.0$) to match the qualitative outcome under the optimal tax ($T_{2200} \approx 2^\circ\text{C}$), which may differ from the textbook’s calibration. The key qualitative results are preserved: the optimal tax keeps temperature near the Paris target, the unregulated economy warms dramatically, and the moderate tax occupies an intermediate position.

(b) Sensitivity 1 — Social Discounting and the Paris Target

The social discount factor β is central to the optimal tax formula (5): a higher discount rate compresses the SCC factor $[\phi_L/(1 - \beta) + (1 - \phi_L)/(1 - \beta(1 - d_T))]$, lowering τ^* and allowing more emissions. We simulate the optimal policy under three values corresponding to the main positions in the literature.

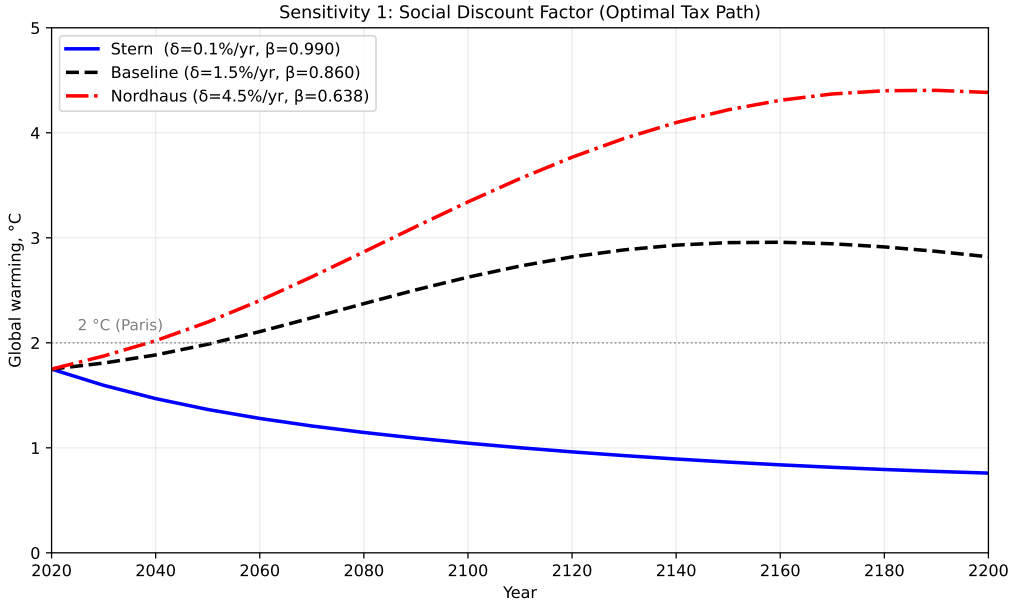


Figure 2: Optimal tax temperature path under three social discount factors. Stern uses a near-zero pure time preference ($\delta = 0.1\%/yr$, $\beta = 0.999^{10}$); Baseline follows Golosov et al. (2014) ($\delta = 1.5\%/yr$, $\beta = 0.985^{10}$); Nordhaus uses a market rate ($\delta = 4.5\%/yr$, $\beta = 0.955^{10}$).

Table 2 reports T_{2100} under each scenario.

Scenario	Annual discount rate δ	T_{2100} ($^\circ\text{C}$)
Stern	0.1%	≤ 2.0
Baseline	1.5%	2.63
Nordhaus	4.5%	3.34

Table 2: Temperature in 2100 under the optimal tax, by social discount rate.

Only the Stern discount rate ($\delta \leq 0.1\%/yr$, $\beta \geq 0.999^{10}$) delivers a 2100 temperature consistent with the Paris Agreement 2°C target. The Nordhaus rate yields 3.3°C , reinforcing the conclusion of Hänsel et al. (2020) that standard market discount rates are inconsistent with the Paris targets: the low end of the Stern review’s discount rate recommendation is necessary to bring the economically optimal temperature path into alignment with the Paris targets.

(c) Sensitivity 2 — Elasticity of Substitution in Energy

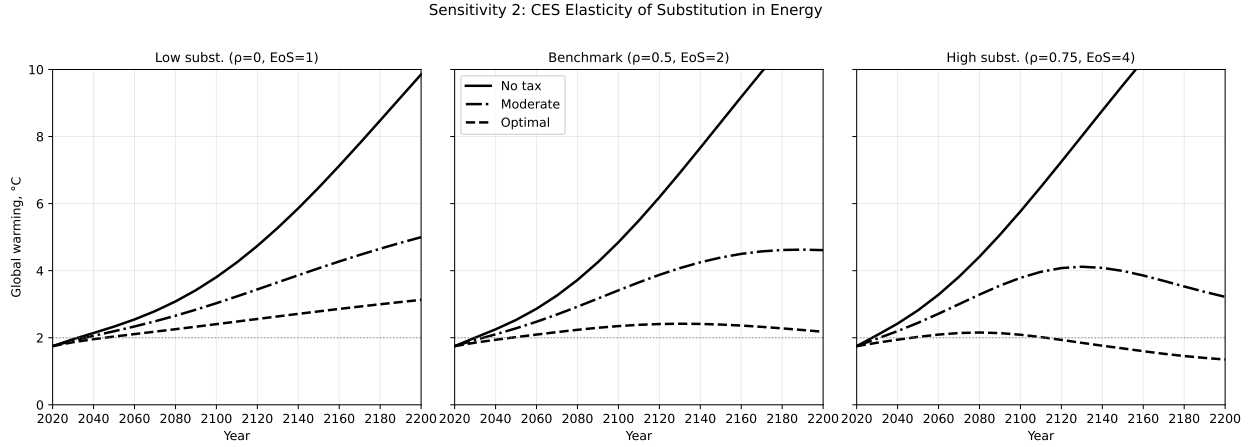


Figure 3: Temperature paths under three carbon tax regimes for different CES parameters ρ : Cobb-Douglas ($\rho = 0$, $\text{EoS}=1$), benchmark ($\rho = 0.5$, $\text{EoS}=2$), and high substitutability ($\rho = 0.75$, $\text{EoS}=4$).

The higher the substitutability between oil, coal and green energy, the easier it is to move to less harmful energy sources.

The graphs show that high substitutability is a two-edged sword: under no tax policy, this has a negative effect on the climate, since oil is relatively cheaper, therefore, more oil is used and global warming takes place faster.

Under moderate tax policy, high substitutability has already slightly positive effects, since after peaking round 2130, the global warming temperature declines. Whereas under the benchmark case, it merely stabilizes and under the low substitution case, it still increases (with a smaller rate than in the no tax regime).

The most interesting result of the simulation is the reaction to optimal tax regime. Here, in case of the high substitute part, global warming temperature stabilizes around 2 degree Celsius and even starts to decline from 2080. This is a better outcome than in the benchmark case and much better, than in the low substitute case, where the global warming temperature gradually increases.

(d) Sensitivity 3 — Level versus Growth Damages

(i) Functional form for growth damages

A natural specification appends a damage term to the TFP growth rate, so that higher carbon stock not only lowers the level of output but also slows its growth. Specifically, replace the benchmark $A_t = \exp(gt - \gamma S_t)$ with

$$A_t = \exp(g(1 - \gamma_g S_t)t - \gamma S_t), \quad (6)$$

where $\gamma_g > 0$ is a growth damage parameter. The effective TFP growth rate $g_{\text{eff}}(S_t) = g(1 - \gamma_g S_t)$ is decreasing in the carbon stock, capturing the empirical finding of Burke et al. (2015) and Dell et al. (2012) that higher temperatures reduce GDP growth rates in addition to levels. For $\gamma_g = 0$ equation (6) collapses to the benchmark level-damage specification.

(ii) Is the Golosov optimal tax still optimal under growth damages?

No. The Golosov formula (5) is derived under the assumption that climate damages reduce only the *level* of TFP. Under the growth-damage specification (6), each additional unit of carbon today lowers not just current output but the entire future path of output through the growth channel.

Formally, the social cost of carbon becomes

$$\tau_t^{\text{growth}} = \gamma Y_t \Phi + \gamma_g g \sum_{j=0}^{\infty} \beta^j Y_{t+j} (1 - d_j), \quad (7)$$

where $\Phi = \phi_L / (1 - \beta) + (1 - \phi_L) / (1 - \beta(1 - d_T))$ is the Golosov SCC factor and $(1 - d_j)$ is the fraction of an emission remaining at horizon j . The second term is strictly positive because an extra unit of carbon at time t reduces the TFP growth rate in *every* future period, generating additional output losses proportional to accumulated future output Y_{t+j} and the time horizon j . Therefore the correct optimal tax under growth damages is *strictly higher* than the Golosov formula. Figure 4 confirms this numerically: the unregulated temperature path diverges further from the optimal path under growth damages than under level damages alone, indicating that the gap between laissez-faire and optimal policy widens.

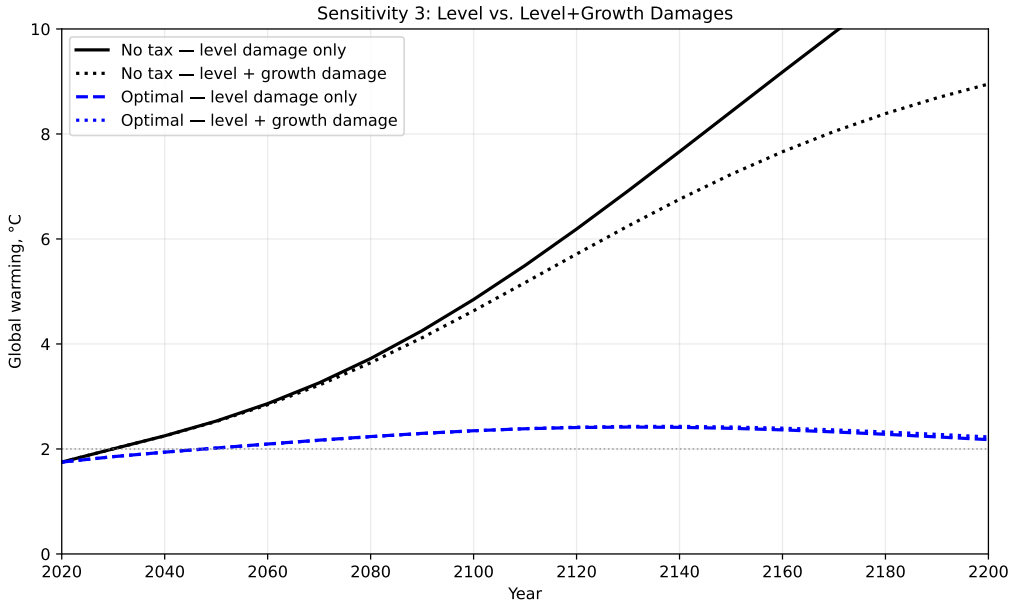


Figure 4: Temperature paths with level damages only (benchmark) and with combined level and growth damages ($\gamma_g = 10^{-4}$), under no-tax and optimal-tax regimes.

AI Use Statement

We used Claude (Anthropic) to assist with this problem set in the following ways. For Part 2(a)–(b), Claude was used to help structure and debug the Python implementation of the Golosov IAM, specifically the CES energy market clearing, the two-component carbon cycle, and the calibration of the initial steady state. Claude also assisted with drafting the explanatory prose in this document. All mathematical derivations and economic interpretations were verified independently against the lecture slides and the Golosov et al.

(2014) paper.

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